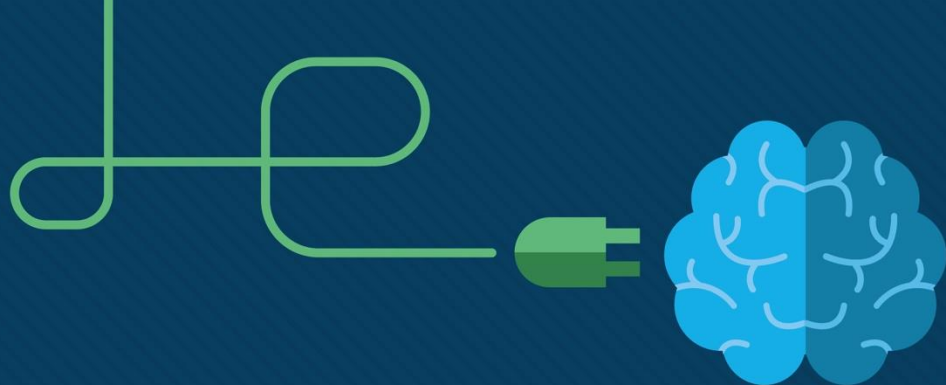




Chapter 3: STP

CCNA Routing and Switching

Scaling Networks v6.0



Chapter 3 - Sections & Objectives

- 3.1 Spanning Tree Concepts
 - Build a simple switched network with redundant links.
 - Explain common problems in a redundant, switched network.
 - Build a simple, switched network using STP.
- 3.2 Varieties of Spanning Tree Protocols
 - Explain how different varieties of spanning tree protocols operate.
 - Describe the different spanning tree varieties.
 - Explain how PVST+ operates.
 - Explain how Rapid PVST+ operates.

Chapter 3 - Sections & Objectives (Cont.)

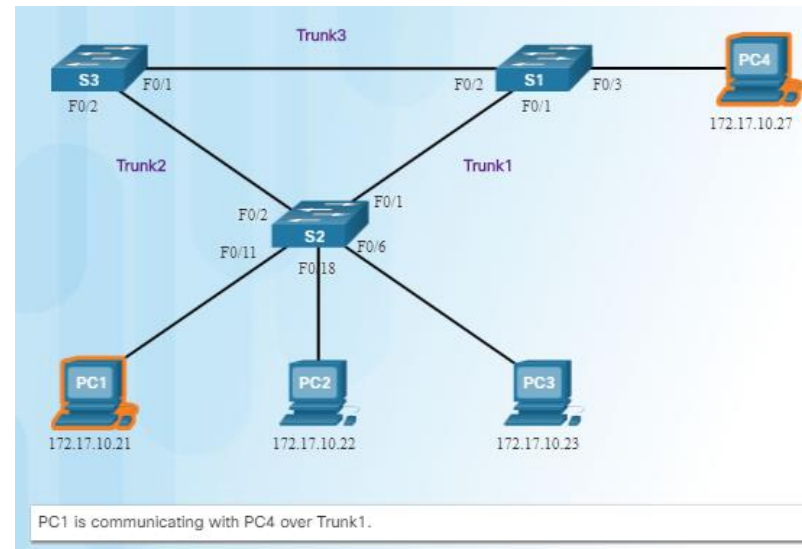
- 3.3 Spanning Tree Configuration
 - Implement PVST+ and Rapid PVST+ in a switched LAN environment.
 - Configure PVST+ in a switched LAN environment.
 - Configure Rapid PVST+ in a switched LAN environment.
 - Analyze common STP configuration issues.
- 3.4 Spanning Tree Key Points
- 3.5 Spanning Tree Assessment

3.1 STP Operation

Spanning Tree

Redundancy at OSI Layers 1 and 2

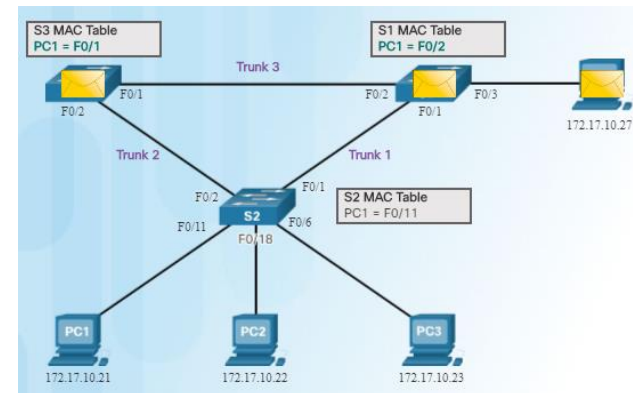
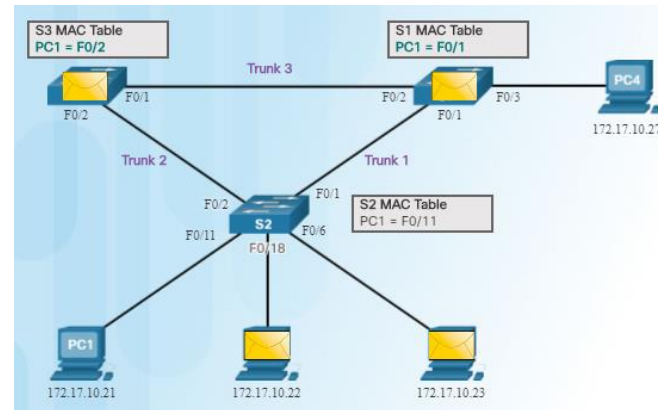
- Switched networks commonly have **redundant paths** and even redundant links between the same two devices.
 - Redundant paths eliminate a **single point of failure** in order to improve reliability and availability.
 - Redundant paths can cause physical and logical Layer 2 **loops**.
- Spanning Tree Protocol (STP)** is a **Layer 2** protocol that helps especially when there are **redundant links**.
- Layer 2 **loop issues**
 - Mac database instability** – copies of the same frame being received on different ports.
 - Broadcast storms** – broadcasts are flooded endlessly causing network disruption.
 - Multiple frame transmission** – multiple copies of unicast frames delivered to the same destination.



Issues with Layer 1 Redundancy: MAC Database Instability

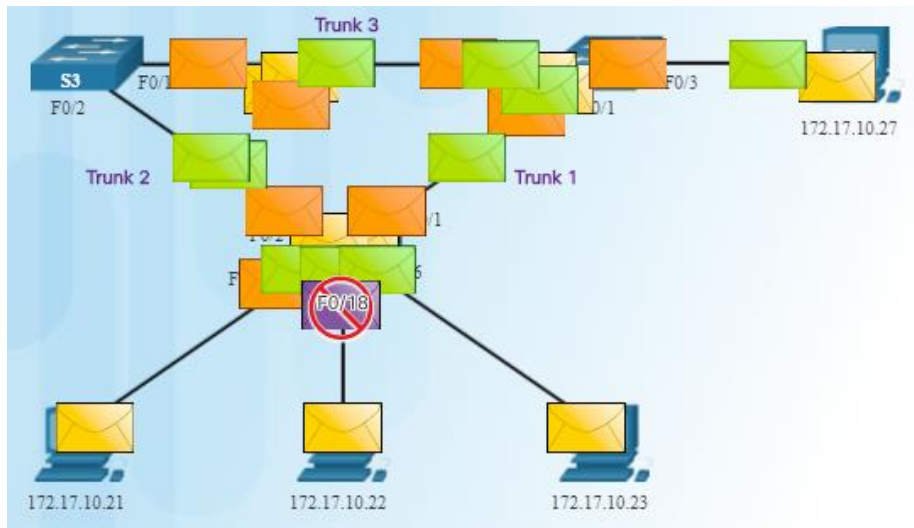
- Ethernet frames do not have a time to live (TTL) field like the Layer 3 IP header has. This means that Ethernet has **no mechanism to drop frames that propagate endlessly**. This can result in **MAC database instability**.

1. PC1 sends a broadcast frame to S2.
 2. S2 updates the MAC address table for PC1's MAC address on port 11.
 3. S2 forwards the frame out all ports except the port the frame came in on. S1 and S3 receive the frame on a trunk and update their own MAC address tables that PC1 is reachable through the trunk port.
 4. S1 and S3 send the frame out all ports except the port it came in on.
 5. When S1 sends the frame out port 2 (Trunk 3), S3 updates the MAC address table to reflect that PC1 is now reachable through port 1.
- A host caught in a network loop is not accessible to other hosts.
 - Due to constant changes in the MAC address table, Switches S3 and S1 do not know which port to forward frames.

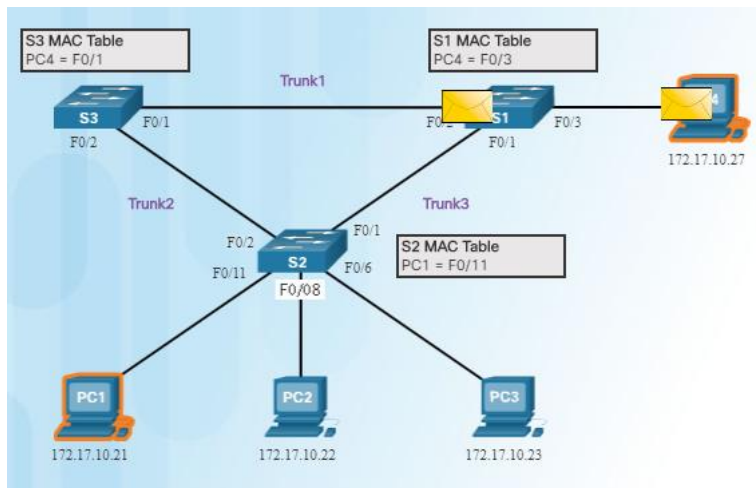


Issues with Layer 1 Redundancy: Broadcast Storms

- Broadcast storm – so many **broadcast frames in a Layer 2 loop** that **use all available bandwidth** and make the network unreachable for legitimate network traffic.
 - Causes a **denial of service (DoS)**
 - Can **develop in seconds** and bring the network down



Issues with Layer 1 Redundancy: Duplicate Unicast Frames



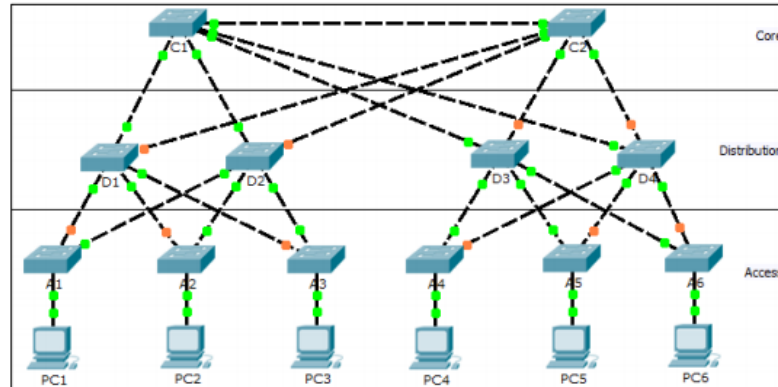
- An unknown **unicast frame** is when the switch does **not** have the **destination MAC address** in its MAC address table and has to **broadcast** the frame out all ports except the port the frame was received on (the ingress port).
- Unknown unicast frames sent onto a **looped network** can result in **duplicate frames** arriving at the destination device.
 1. PC1 sends a frame destined for PC4.
 2. S2 does not have PC4's MAC address in the MAC address table so it forwards the frame out all ports including the trunks that lead to S1 and S3. S1 sends the frame to PC4. S3 also sends a copy of the frame over to S1 which delivers the same frame again to PC4.

Packet Tracer – Examining a Redundant Design



Packet Tracer – Examining a Redundant Design

Topology



Objectives

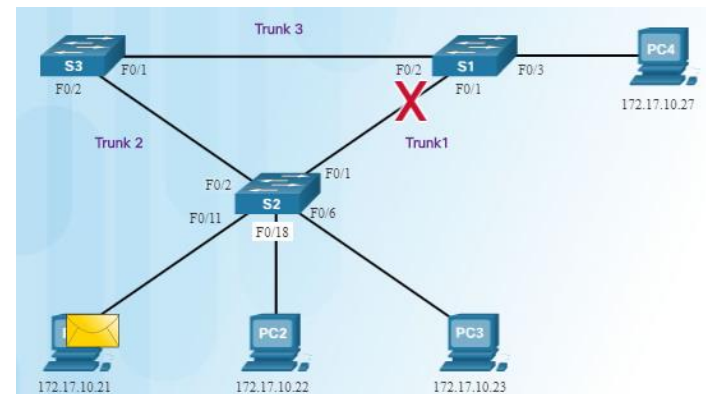
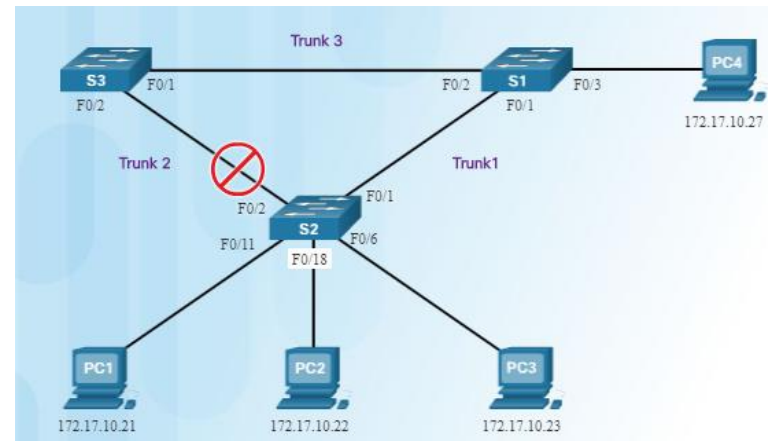
Part 1: Check for STP Convergence

Part 2: Examine the ARP Process

Part 3: Test Redundancy in a Switched Network

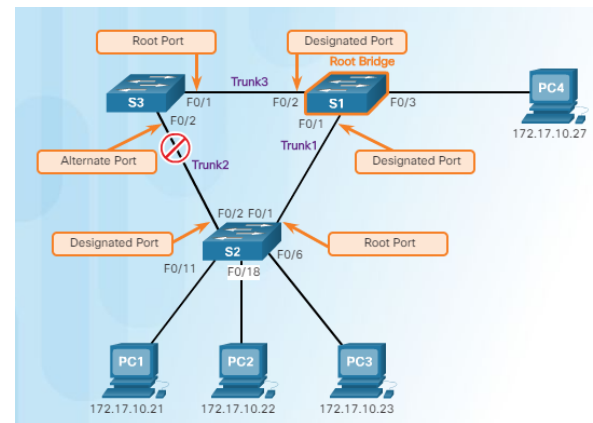
Spanning Tree Algorithm: Introduction

- The Spanning Tree Protocol (STP) creates **one logical path through the switch network** (all destinations on the network).
- **Blocks redundant paths** that could cause a loop.
- STP sends **bridge protocol data units (BPDUs)** between Layer 2 devices in order to create the one logical path.
- A port on S2 is blocked so traffic can only flow one way between any two devices.
- When **Trunk1 fails**, the blocked port on S2 is **unblocked** and traffic can flow between S2 and S3.



Spanning Tree Algorithm: Port Roles

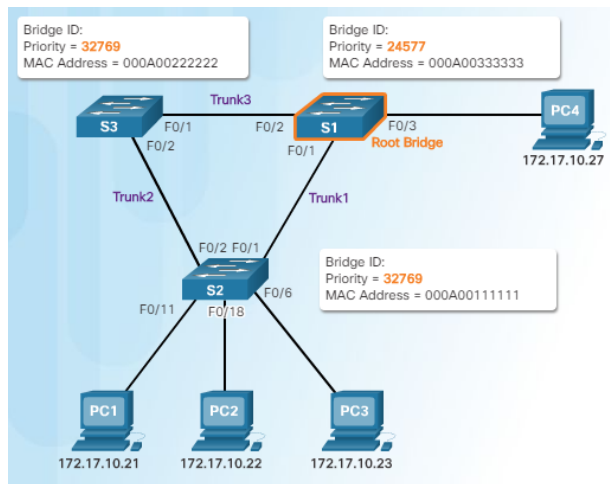
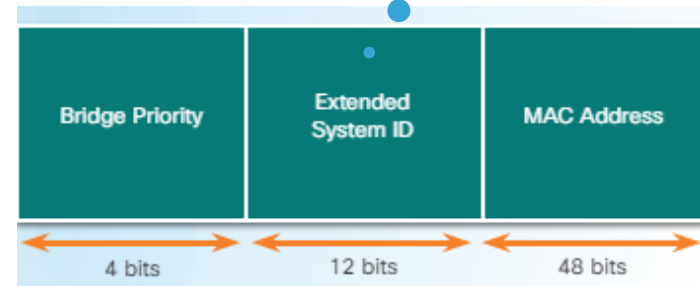
- **Root bridge** – one Layer 2 device in a switched network.
- **Root port** – one port on a switch that has the lowest cost to reach the root bridge.
- **Designated port** – selected on a per-segment (each link) basis, based on the cost to get back to root bridge for either side of the link.
- **Alternate port** – (RSTP only) backup port for the designated port when the other side is not a root port.
- **Backup port** – (RSTP only) backup port for the root port.



Spanning Tree Algorithm: Root Bridge

Supports per-VLAN STP operations

- **Lowest bridge ID (BID)** becomes root bridge
 - Originally BID had **two fields**: **bridge priority** and **MAC address**
 - Bridge priority default is **32,768** (can change)
 - **Lowest MAC address** (if bridge priority is not changed) becomes determinant for root bridge.

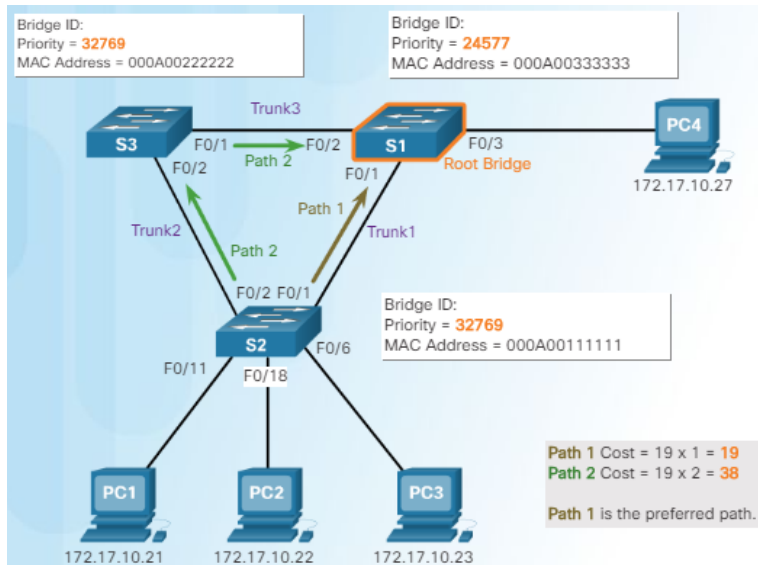


STP Operation

Spanning Tree Algorithm: Root Path Cost

- Root path cost is **used to determine the role of the port** and whether or not traffic is blocked.
- Can be modified with the **spanning-tree cost interface command**.

Link Speed	Cost (Revised IEEE Specification)	Cost (Previous IEEE Specification)
10 Gb/s	2	1
1 Gb/s	4	1
100 Mb/s	19	10
10 Mb/s	100	100



```
S2# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S2(config)# interface f0/1
S2(config-if)# spanning-tree cost 25
```

```
S2(config)# interface f0/1
S2(config-if)# no spanning-tree cost
```

```
S2# show spanning-tree

VLAN001
  Spanning tree enabled protocol ieee
  Root ID    Priority 24577
            Address 000A.0033.3333
            Cost   19
            Port   1
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

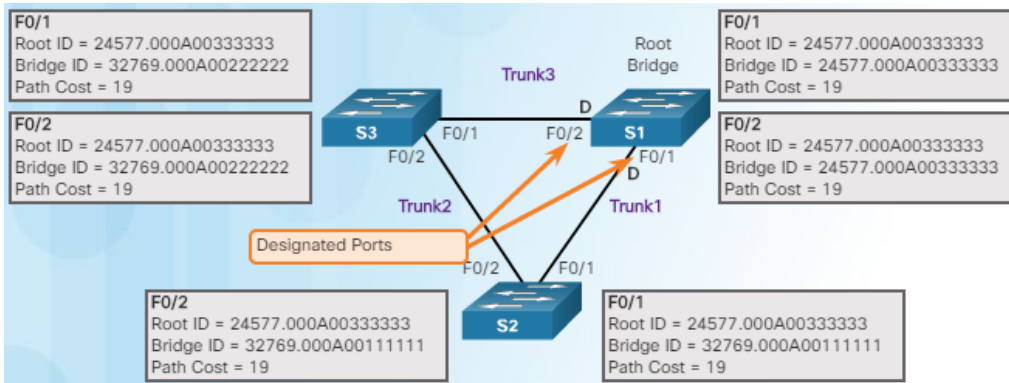
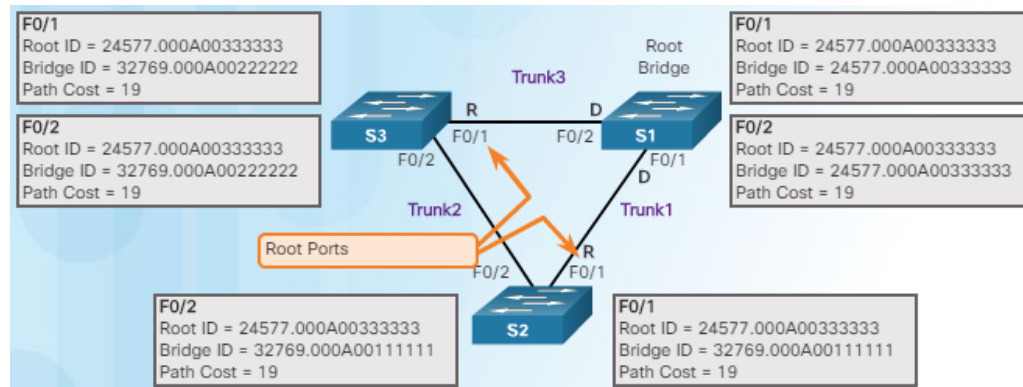
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address 000A.0011.1111
            Hello time 2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time 300

Interface Role Sts Cost Prio.Nbr Type
-----
F0/1      Root FWD 19    128.1   Edge P2p
F0/2      Desg FWD 19    128.2   Edge P2p
```

STP Operation

Port Role Decisions for RSTP

- S1 is root bridge

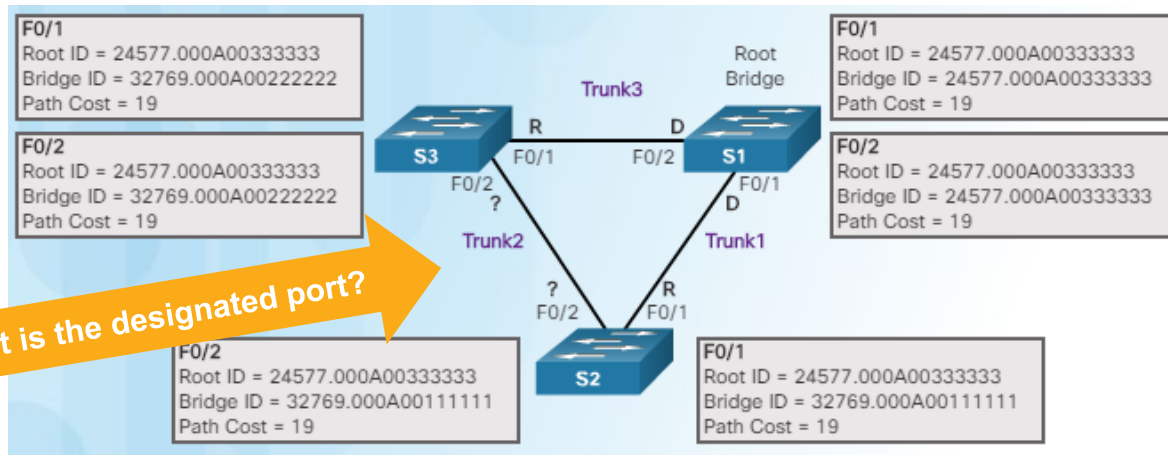


STP Operation

Port Role Decisions for RSTP (Cont.)

Which switch (S3 or S2) has the lowest BID?

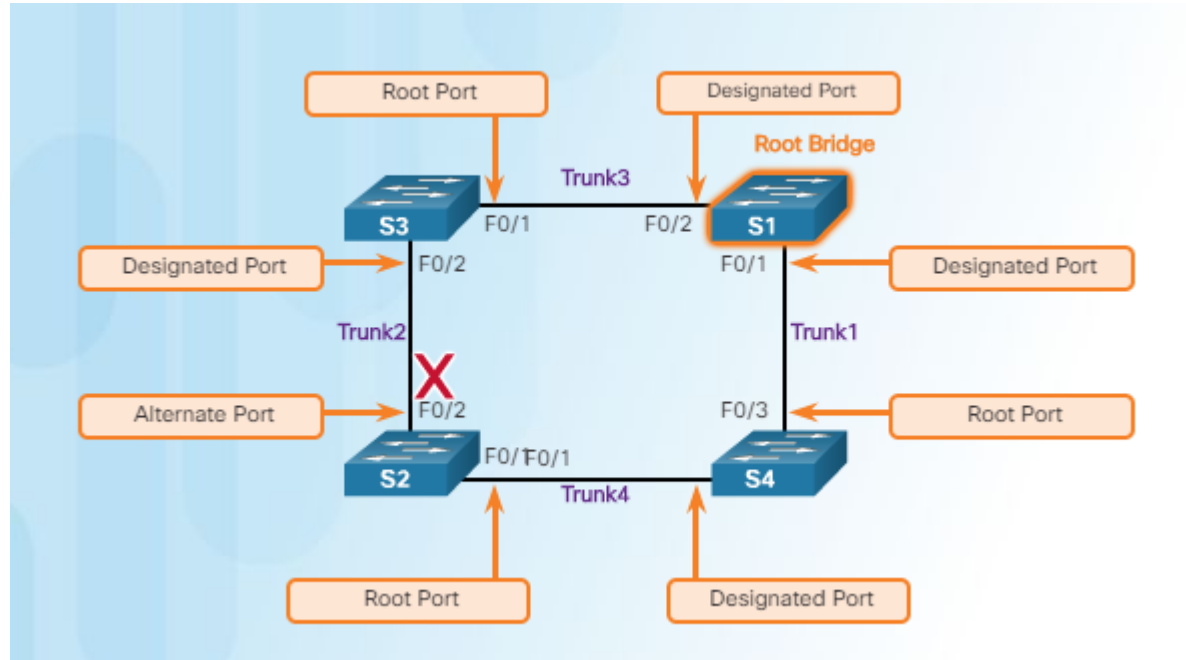
Which port is the designated port?



- After S3 and S2 exchange BPDUs, STP determines that the F0/2 port on S2 becomes the designated port and the S3 F0/2 port becomes the alternate port, thus going into the blocking state so there is only one path through the switched network.

STP Operation

Determine Designated and Alternate Ports



Remember port states are based on path cost back to root bridge.

STP Operation

802.1D BPDU Frame Format

Field Number	Bytes	Field
1-4	2	Protocol ID
	1	Version
	1	Message Type
	1	Flags
5-8	8	Root ID
	4	Root Path Cost
	8	Bridge ID
	2	Port ID
9-12	2	Message Age
	2	Max Age
	2	Hello Time
	2	Forward Delay

```

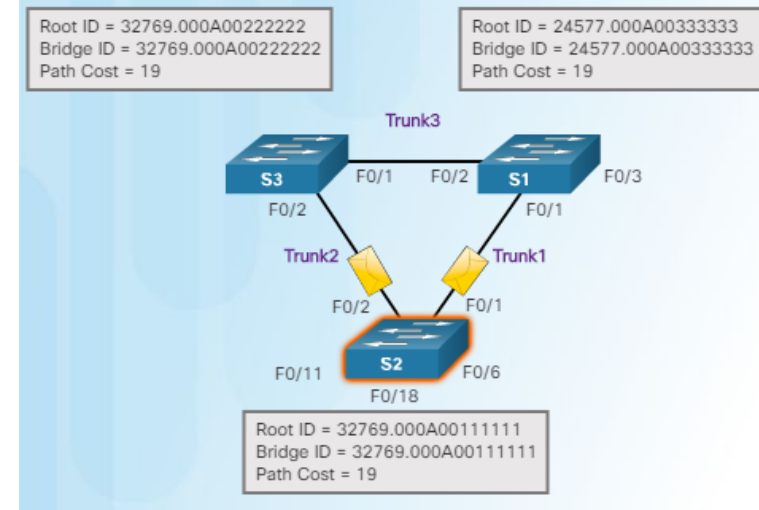
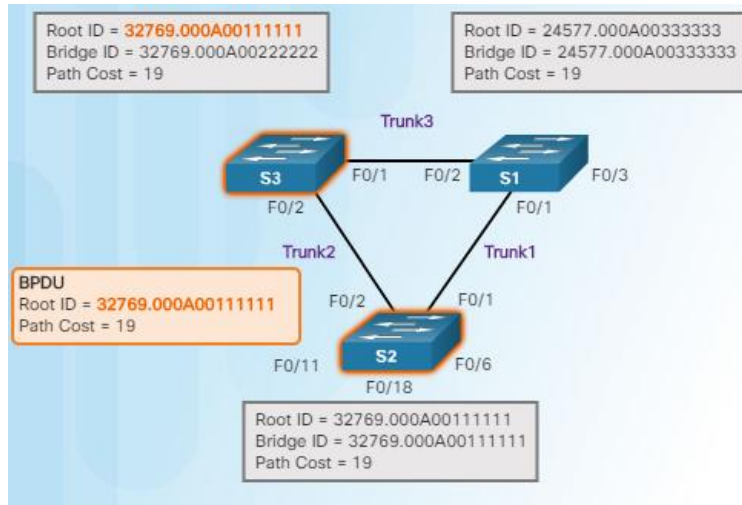
Frame 1 (60 bytes on wire, 60 bytes captured)
  IEEE 802.3 Ethernet
    Destination: Spanning-tree-(for-bridges)_00 (01:80:c2:00:00:00)
    Source: Cisco_9e:93:03 (00:19:aa:9e:93:03)
    Length: 38
    Trailer: 0000000000000000
  Logical-Link Control
    Spanning Tree Protocol
      Protocol Identifier: Spanning Tree Protocol (0x0000)
      Protocol version Identifier: Spanning Tree (0)
      BPDU Type: Configuration (0x00)
      BPDU flags: 0x01 (Topology change)
      Root Identifier: 24577 / 00:19:aa:9e:93:00
      Root Path Cost: 0
      Bridge Identifier: 24577 / 00:19:aa:9e:93:00
      Port Identifier: 0x8003
      Message Age: 0
      Max Age: 20
      Hello Time: 2
      Forward Delay: 15
  
```

Field	Description
Protocol ID	Type of protocol being used; set to 0
Version	Protocol version; set to 0
Message type	Type of message; set to 0
Flags	Topology change (TC) bit signals a topology a change; topology change acknowledgment (TCA) bit used when a configuration message with the TC bit set has been received
Root ID	Root bridge information
Root path cost	Cost of the path from the switch sending the configuration message to the root bridge
Bridge ID	Includes priority, extended system ID, and MAC address ID of the bridge sending the message
Port ID	Port number from which the BPDU was sent
Message age	Amount of time since the root bridge sent the configuration message
Max age	When the current configuration message will be deleted
Hello time	Time between root bridge messages
Forward delay	Time the bridges should wait before going to a new state

STP Operation

802.1D BPDUs Propagation and Process

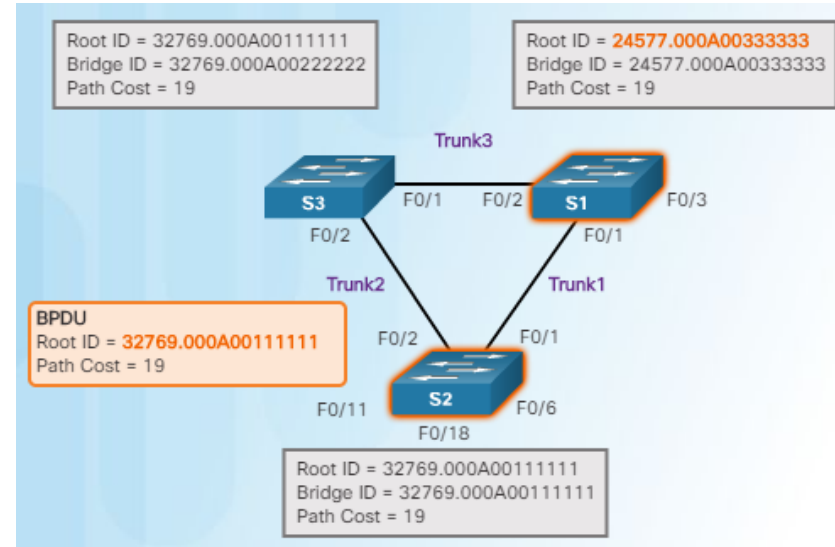
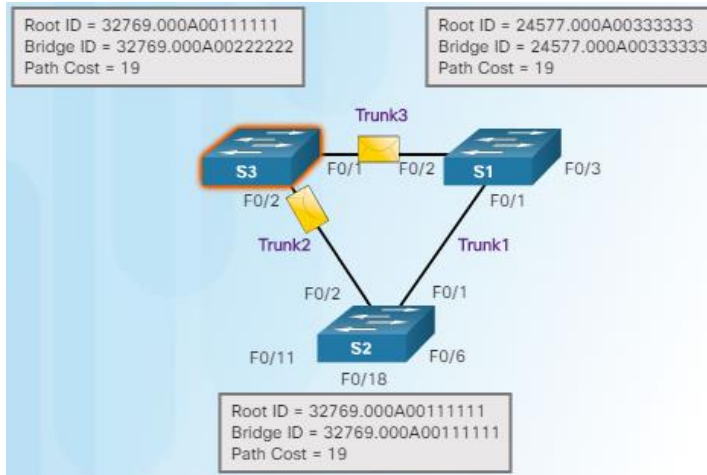
1. When a switch is powered on, it assumes it is the root bridge until BPDUs are sent and STP calculations are performed. S2 sends out BPDUs.
2. S3 compares its root ID with the BPDUs from S2. S2 is lower so S3 updates its root ID.



STP Operation

802.1D BPDU Propagation and Process (Cont.)

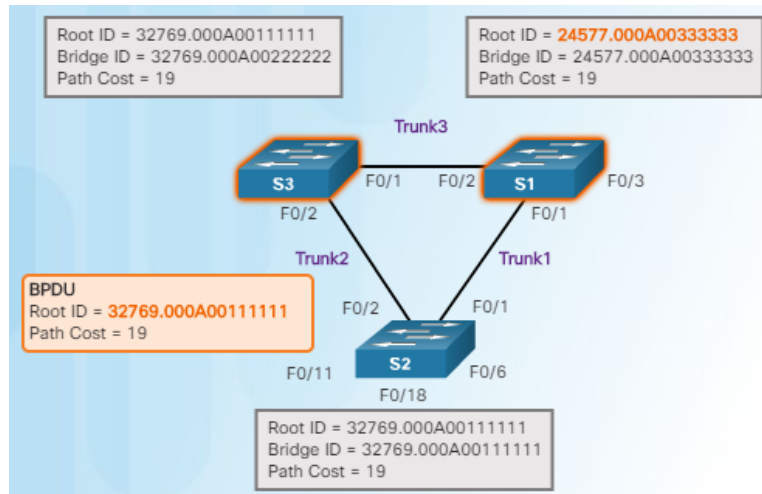
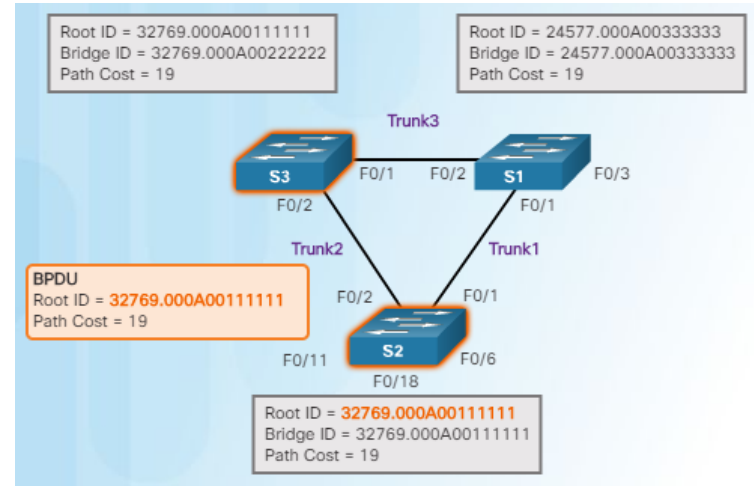
3. S1 receives the same information from S2 and because S1 has a lower BID, it ignores the information from S2.
4. S3 sends BPDUs out all ports indicating that S2 is root bridge.



STP Operation

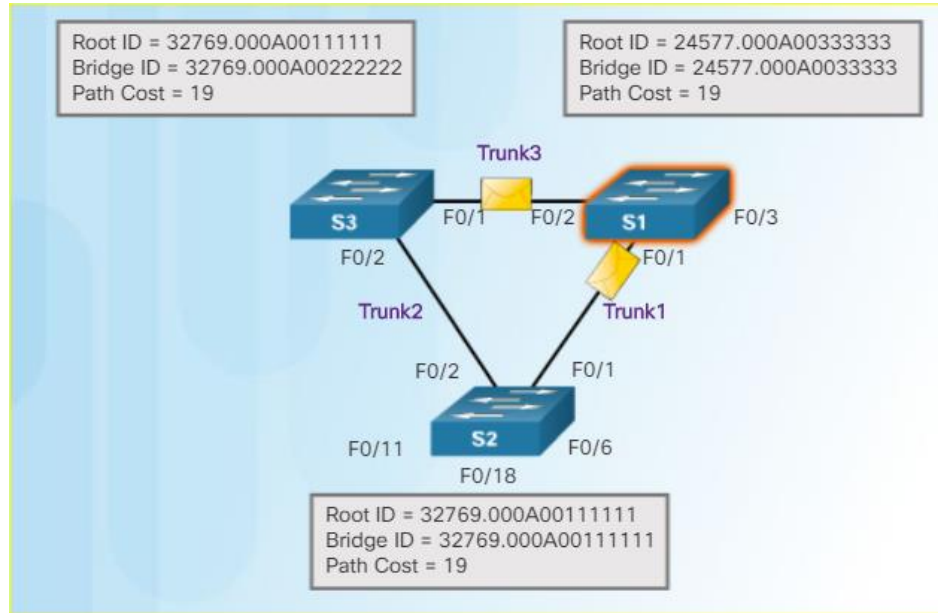
802.1D BPDUs Propagation and Process (Cont.)

5. S2 compares the info from S3 so S2 still thinks it is root bridge.
6. S1 gets the same information from S3 (that S2 is root bridge), but because S1 has a lower BID, the switch ignores the information in the BPDUs.



802.1D BPDUs Propagation and Process (Cont.)

7. S1 now sends out BPDUs out all ports. The BPDU contains information designated S1 as root bridge.

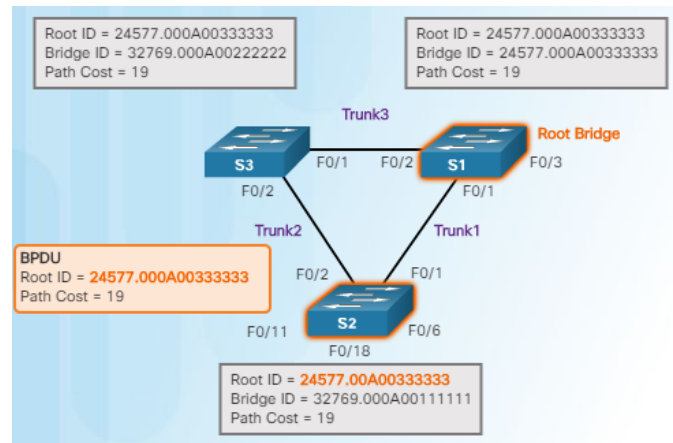
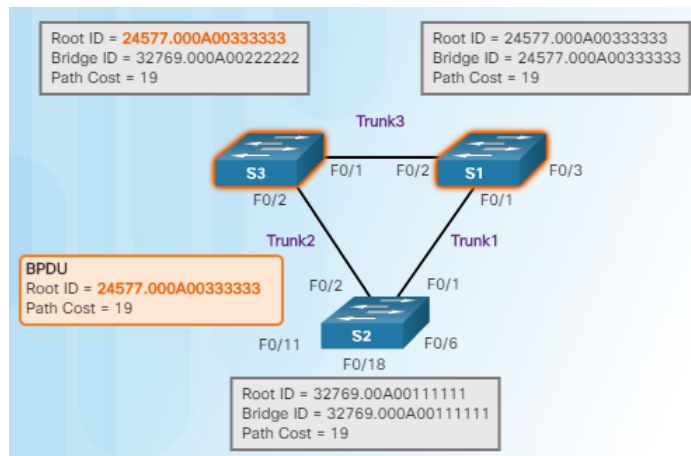


STP Operation

802.1D BPDUs Propagation and Process (Cont.)

8. S3 compares the info from S1 so S3 now sees that the BID from S1 is lower than its stored root bridge information which is currently showing that S2 is root bridge. S3 changes the root ID to the information received from S1.
9. S2 compares the info from S1 so S2 now sees the BID from S1 is lower than its own BID. S2 now updates its own information showing S1 as root bridge.

Remember that after root bridge has been determined, the other port roles can be determined because those roles are determined by total path cost back to root bridge.

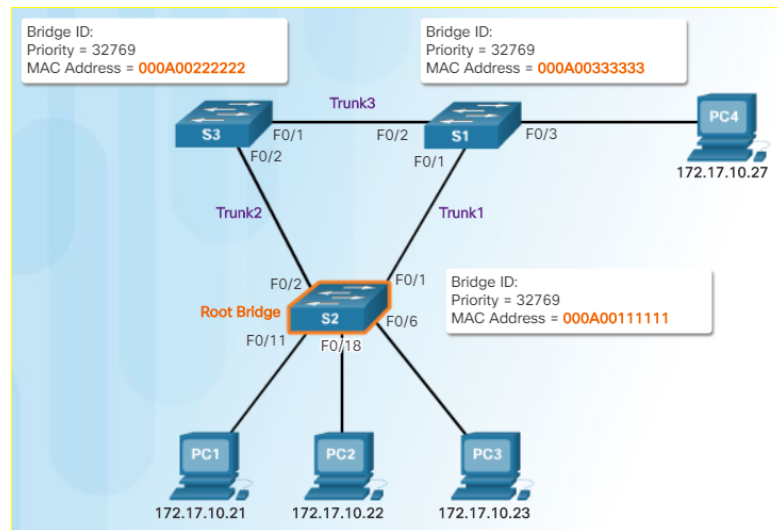
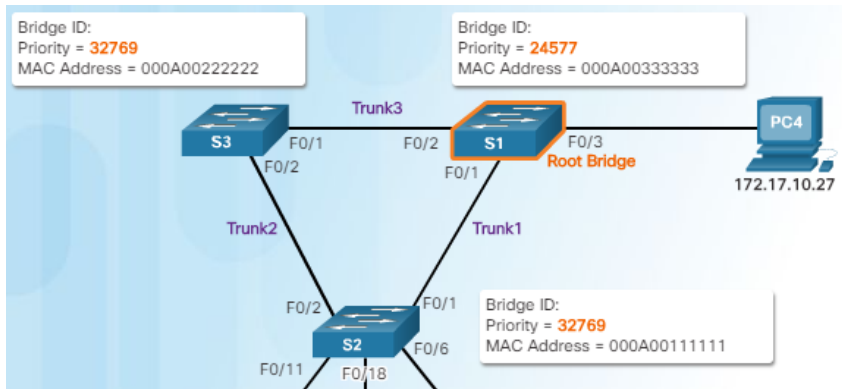
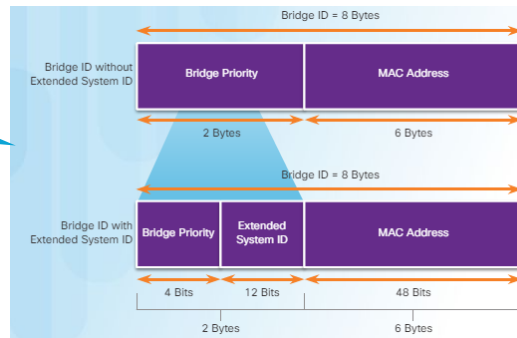


STP Operation

Extended System ID

Remember -
lowest BID
becomes root

- If priorities are all set to the default, **lowest MAC address** is the determining factor in lowest BID.
- The **priority value** can be modified to **influence root bridge elections**.



Video Demonstration – Observing Spanning Tree Protocol Operation

Cisco Networking Academy

Observing Spanning Tree Protocol Operation

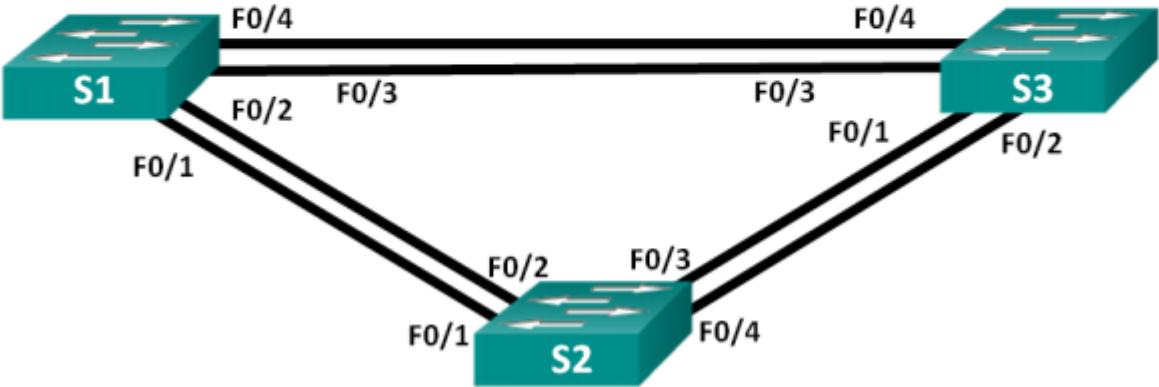
Video Objective

- STP Characteristics
- Debugging STP
- Observing Fault Tolerance with STP
- Observing a Broadcast Storm without STP

Building a Switched Network with Redundant Links

Lab – Building a Switched Network with Redundant Links

Topology



3.2 Types of Spanning Tree Protocols

Types of Spanning Tree Protocols

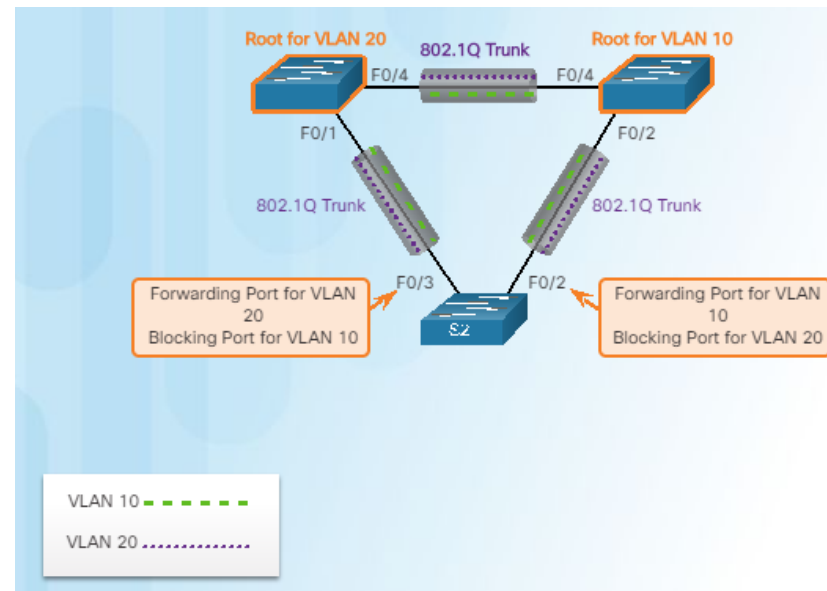
STP Type	Description
802.1D	1998 - Original STP standard
CST	One spanning-tree instance
PVST+	Cisco update to 802.1D; each VLAN has its own spanning-tree instance
802.1D	2004 – Updated bridging and STP standard
802.1w (RSTP)	Improves convergence by adding new roles to ports and enhancing BPDU exchange
Rapid PVST+	Cisco enhancement of RSTP using PVST+
802.1s (MSTP)	Multiple VLANs can have the same spanning-tree instance

Characteristics of Spanning Tree Protocols

STP Type	Standard	Resources Needed	Convergence	Tree Calculation
STP	802.1D	Low	Slow	All VLANs
PVST+	Cisco	High	Slow	Per VLAN
RSTP	802.1w	Medium	Fast	All VLANs
Rapid PVST+	Cisco	Very high	Fast	Per VLAN
MSTP	802.1s	Medium or high	Fast	Per instance

Overview of PVST+

- Original **802.1D** defines a common spanning tree
 - **One** spanning tree instance for the switched network (no matter how many VLANs)
 - **No load sharing**
 - **One uplink must block for all VLANs**
 - **Low CPU utilization** because only one instance of STP is used/calculated
- **Cisco PVST+** - each VLAN has its own spanning tree instance
 - One port can be **blocking for one VLAN** and **forwarding for another VLAN**
 - Can **load balance**
 - Can stress the **CPU** if a large number of VLANs are used



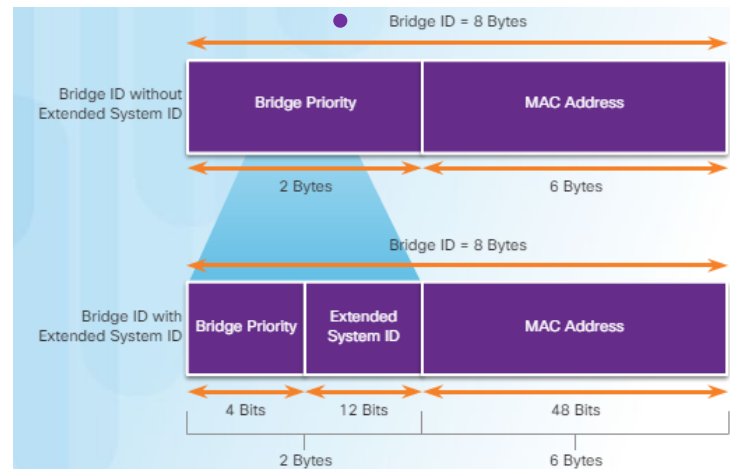
Port States and PVST+ Operation

	Port State				
Operation allowed	Blocking	Listening	Learning	Forwarding	Disabled
Can receive/process BPDUs	Yes	Yes	Yes	Yes	No
Can forward data frames received on an interface	No	No	No	Yes	No
Can forward data frames switched from another interface	No	No	No	Yes	No
Can learn MAC addresses	No	No	Yes	Yes	No

Extended System ID and PVST+ Operation

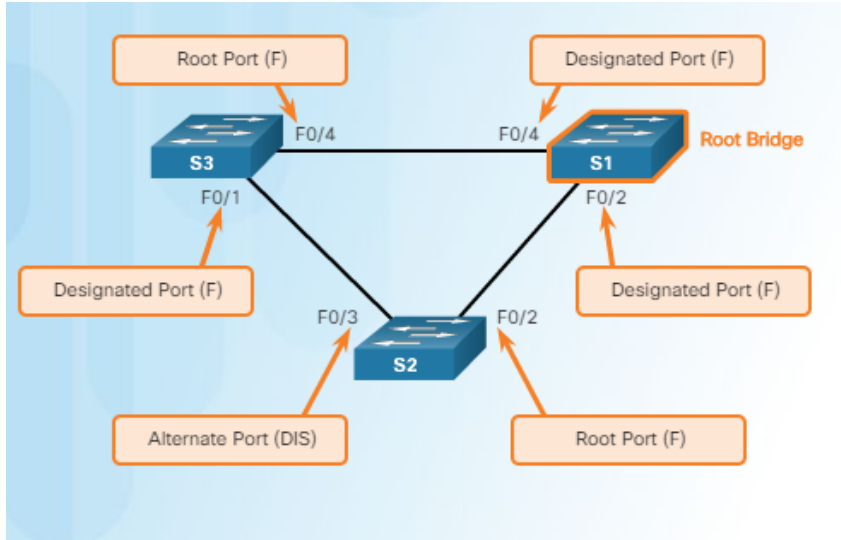
- The extended system ID field ensures each switch has a **unique BID for each VLAN**.
- The VLAN number is added to the priority value.
 - Example – VLAN 2 priority is 32770 (default value of 32768 plus the VLAN number of 2 equals 32770)
 - Can modify the priority number to influence the root bridge decision process
- Reasons to select a particular switch as root bridge
 - Switch is positioned such that most traffic patterns flow toward this particular switch
 - Switch has more processing power (better CPU)
 - Switch is easier to access and manage remotely

Remember that the BID is a unique ID



Varieties of Spanning Tree Protocols

Overview of Rapid PVST+



- Rapid PVST+ speeds up STP recalculations and converges quicker
 - Cisco version of RSTP
- Two new port types
 - Alternate port (DIS)
 - Backup port
- Independent instance of RSTP runs for each VLAN
- Cisco features such as UplinkFast and BackboneFast are not compatible with switches that run RSTP

Varieties of Spanning Tree Protocols

RSTP BPDUs

- RSTP uses type 2, version 2 BPDUs
 - Original version was type 0, version 0
- A switch using **RSTP** can work with and communicate with a switch running the **original 802.1D** version
- BPDUs are used as a **keepalive mechanism**
 - **3 missed BPDUs indicates lost connectivity**

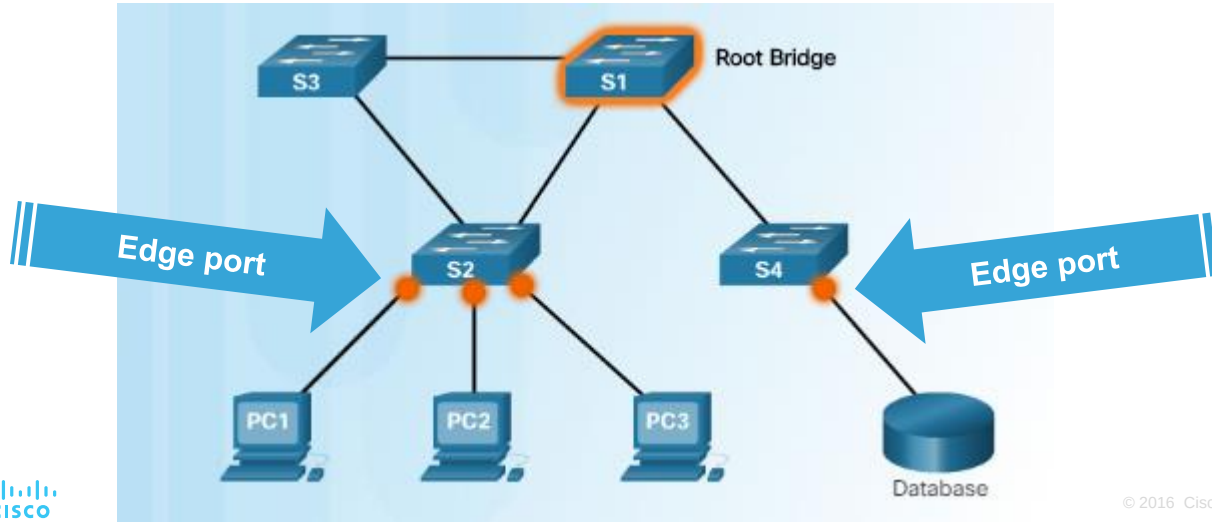
RSTP Version 2 BPDU	
Field	Byte Length
Protocol ID=0x0000	2
Protocol Version ID=0x02	1
BPDU Type=0x02	1
Flags	1
Root ID	8
Root Path Cost	4
Bridge ID	8
Port ID	2
Message Age	2
Max Age	2
Hello Time	2
Forward Delay	2

Flag Field	
Field Bit	Bit
Topology Change	0
Proposal	1
Port Role	2-3
Unknown Port	00
Alternate or Backup Port	01
Root Port	10
Designated Port	11
Learning	4
Forwarding	5
Agreement	6
Topology Change Acknowledgment	7

Varieties of Spanning Tree Protocols

Edge Ports

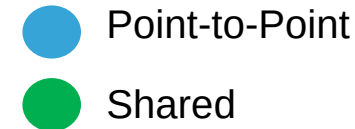
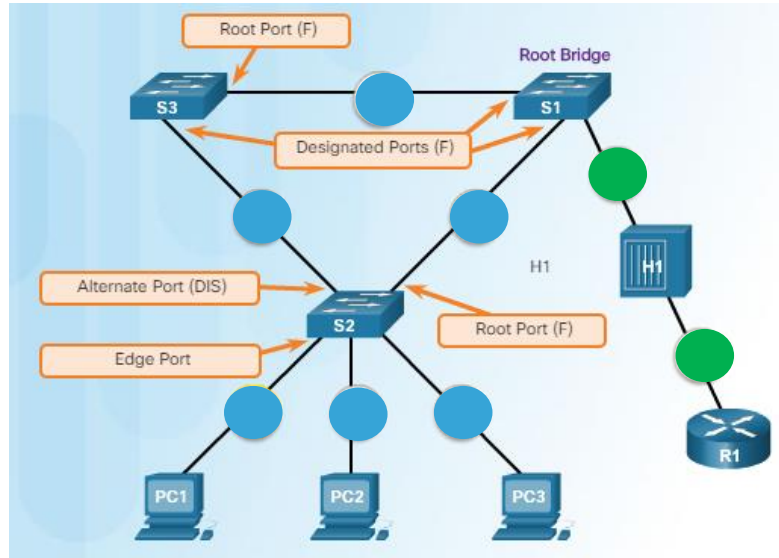
- Has an end device connected – **NEVER** another switch
- Immediately goes to the forwarding state
- Functions similar to a port configured with Cisco PortFast
- Use the **spanning-tree portfast** command



Varieties of Spanning Tree Protocols

Link Types

- Point-to-Point – a port in full-duplex mode connecting from one switch to another switch or from a device to a switch
- Shared – a port in half-duplex mode connecting a hub to a switch



3.3 Spanning Tree Configuration

PVST+ Configuration

Catalyst 2960 Default Configuration

Feature	Default Setting
Enable state	Enabled on VLAN 1
Spanning-tree mode	PVST+ (Rapid PVST+ and MSTP are disabled)
Switch priority	32768
Spanning-tree port priority (configurable on a per-interface basis)	128
Spanning-tree port cost (configurable on a per-interface basis)	1000 Mb/s: 4 100 Mb/s: 19 10 Mb/s: 100
Spanning-tree VLAN port priority (configurable on a per-VLAN basis)	128
Spanning-tree VLAN port cost (configurable on a per-VLAN basis)	1000 Mb/s: 4 100 Mb/s: 19 10 Mb/s: 100
Spanning-tree timers	Hello time: 2 seconds Forward-delay time: 15 seconds Maximum-aging time: 20 seconds Transmit hold count: 6 BPDUs

Configuring and Verifying the Bridge ID

- Two ways to influence the root bridge election process
 - Use the **spanning-tree vlan x root primary** or **secondary** command.
 - Change the priority value by using the **spanning-tree vlan x priority x** command.
- Verify the bridge ID and root bridge election by using the **show spanning-tree** command.

```
S3# show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00A.0033.3333
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
             Address     00A.0033.3333
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 300

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1 Desg FWD 4 128.1 p2p
Fa0/2 Desg FWD 4 128.2 p2p
```

Method 1

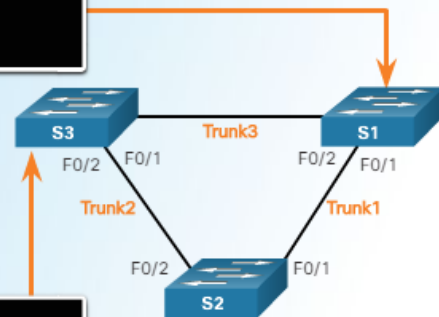
```
s1(config)# spanning-tree VLAN 1 root primary
s1(config)# end
```

Method 2

```
s3(config)# spanning-tree VLAN 1 priority 24576
s3(config)# end
```

Method 1

```
s2(config)# spanning-tree VLAN 1 root secondary
s2(config)# end
```

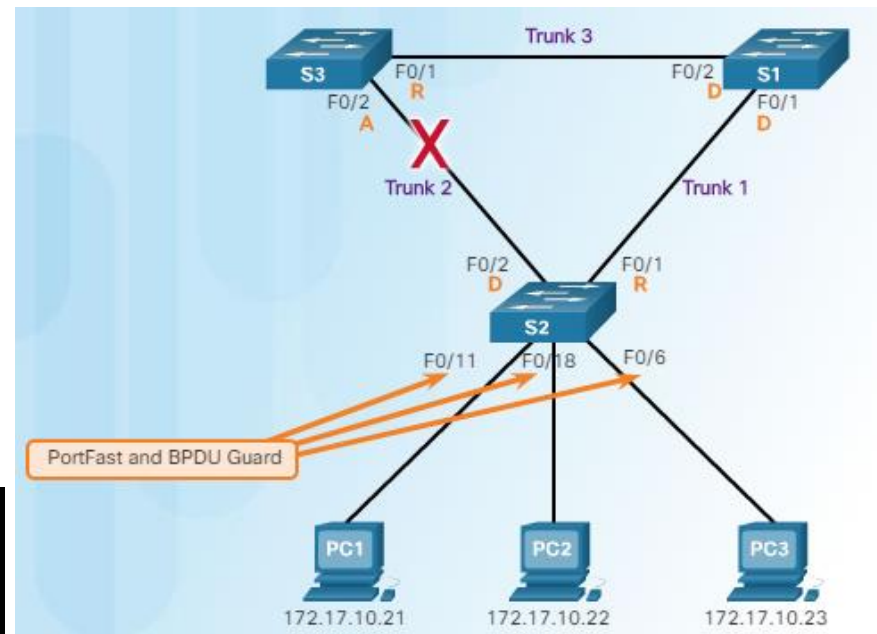


PortFast and BPDU Guard

- **PortFast** is used on ports that have **end devices attached**.
 - Puts a port in the forwarding state
 - Allows DHCP to work properly
- **BPDU Guard** disables a port that has PortFast configured on it if a **BPDU is received**

```
S2(config)# interface FastEthernet 0/11
S2(config-if)# spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single host.
Connecting hubs, concentrators, bridges, etc... to this interface
when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/11 but will only
have effect when the interface is in a non-trunking mode.
S2(config-if)# spanning-tree bpduguard enable
```



```
S2# show running-config interface f0/11
Building configuration...

Current configuration : 90 bytes
!
interface FastEthernet0/11
 spanning-tree portfast
 spanning-tree bpduguard enable
```

PVST+ Configuration

PVST+ Load Balancing

```
S3(config)# spanning-tree vlan 20 root primary
S3(config)# spanning-tree vlan 10 root secondary
```

or

```
S3(config)# spanning-tree vlan 20 priority 4096
```

```
S1(config)# spanning-tree vlan 10 root primary
S1(config)# spanning-tree vlan 20 root secondary
```

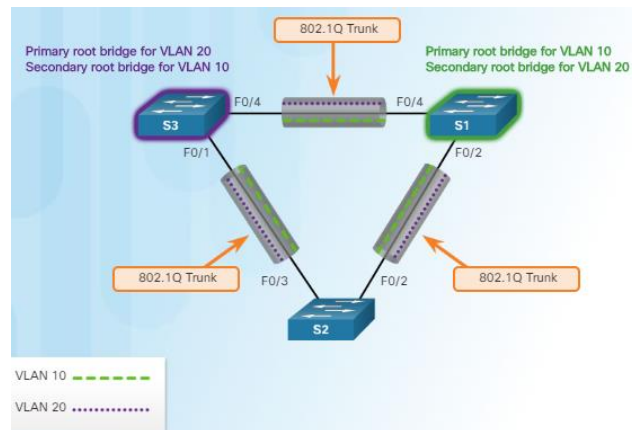
or

```
S1(config)# spanning-tree vlan 10 priority 4096
```

```
S1# show spanning-tree active
<output omitted>

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    4106
             Address     0019.aa9e.b000
             This bridge is the root
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
  Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
             Address     0019.aa9e.b000
             Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
             Aging Time 300

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/2 Desg FWD 19 128.2 p2p
Fa0/4 Desg FWD 19 128.4 p2p
```



```
S1# show running-config
Building configuration...

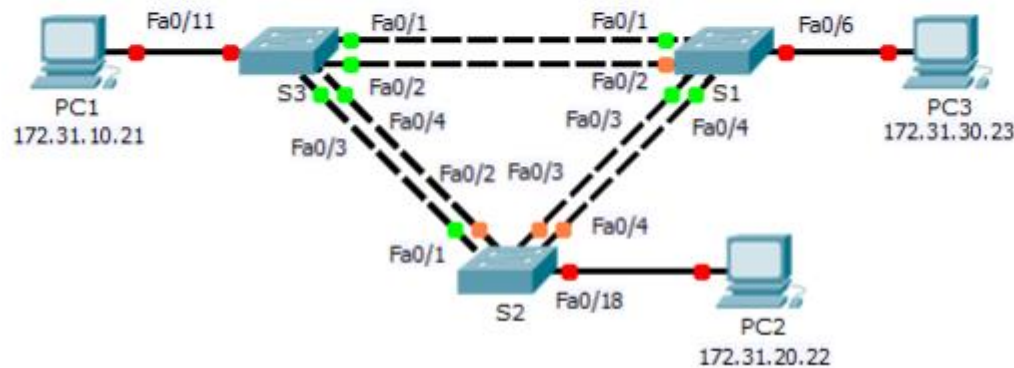
Current configuration : 1595 bytes
!
version 12.2
<output omitted>
!
spanning-tree mode pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 24576
spanning-tree vlan 10 priority 4096
spanning-tree vlan 20 priority 28672
```


PVST+ Configuration

Packet Tracer – Configuring PVST+

Packet Tracer – Configuring PVST+

Topology



Rapid PVST+ Configuration

Spanning Tree Mode

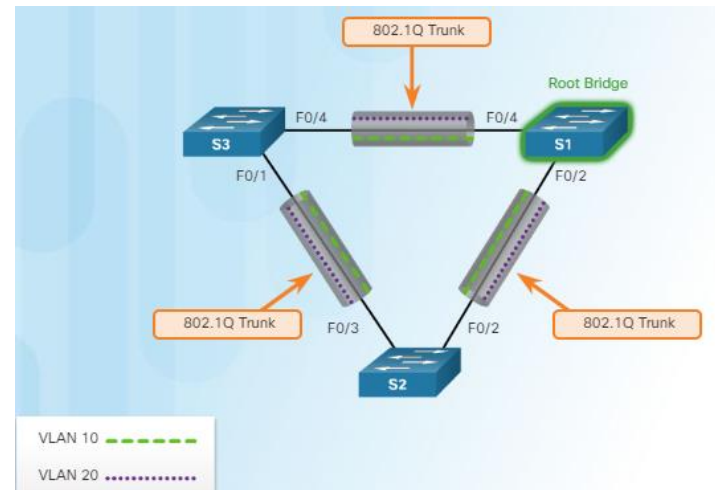
- Rapid PVST+ supports RSTP on a per-VLAN basis.
 - Default on a 2960 is PVST+.
 - The **spanning-tree mode rapid-pvst** puts a switch into Rapid PVST+ mode.
 - The **spanning-tree link-type point-to-point** interface command designates a particular port as a point-to-point link (does not have a hub attached).
 - The **clear spanning-tree detected-protocols** privileged mode command is used to clear STP.

```
S1# configure terminal
S1(config)# spanning-tree mode rapid-pvst
S1(config)# interface f0/2
S1(config-if)# spanning-tree link-type point-to-point
S1(config-if)# end
S1# clear spanning-tree detected-protocols
```

```
S1# show run

<output omitted>

spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 1 priority 24576
spanning-tree vlan 10 priority 4096
spanning-tree vlan 20 priority 28672
```



```
S1# show spanning-tree vlan 10
```

```
VLAN0010
  Spanning tree enabled protocol rstp
    Root ID    Priority    4106
              Address     0019.aa9e.b000
              This bridge is the root
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
    Bridge ID  Priority    4106 (priority 4096 sys-id-ext 10)
              Address     0019.aa9e.b000
              Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
              Aging Time 300

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/2 Desg LRN 19 128.2 P2p
Fa0/4 Desg LRN 19 128.4 P2p
```

Packet Tracer – Configuring Rapid PVST+



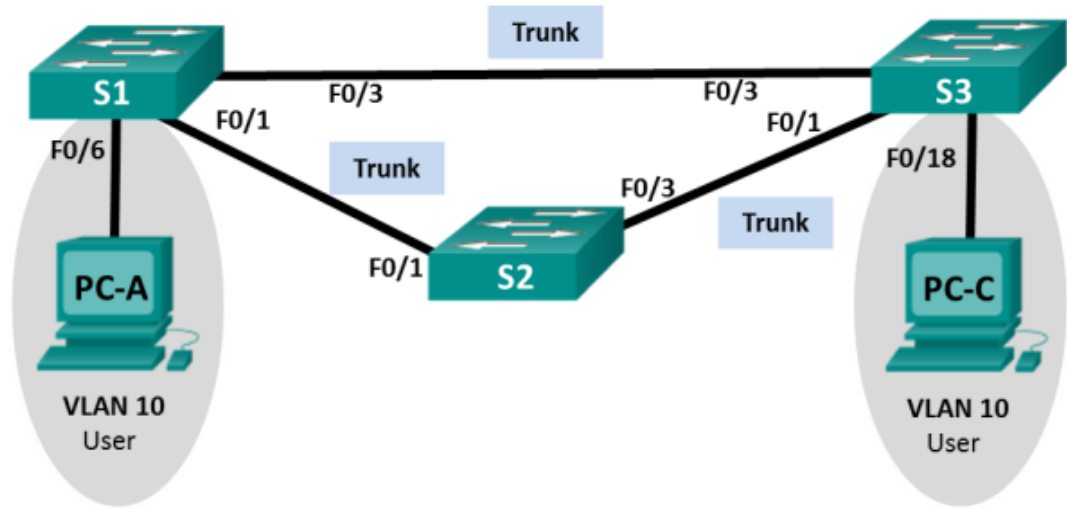
Packet Tracer – Configuring Rapid PVST+

Rapid PVST+ Configuration

Packet Tracer – Configuring Rapid PVST+, PortFast and BPDU Guard

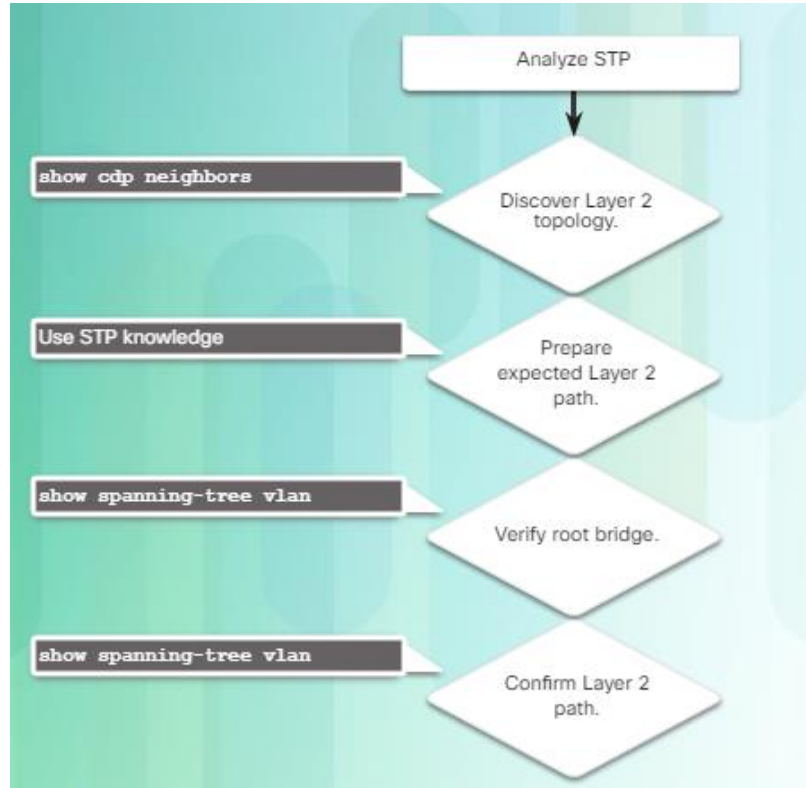
Lab – Configuring Rapid PVST+, PortFast, and BPDU Guard

Topology



STP Configuration Issues

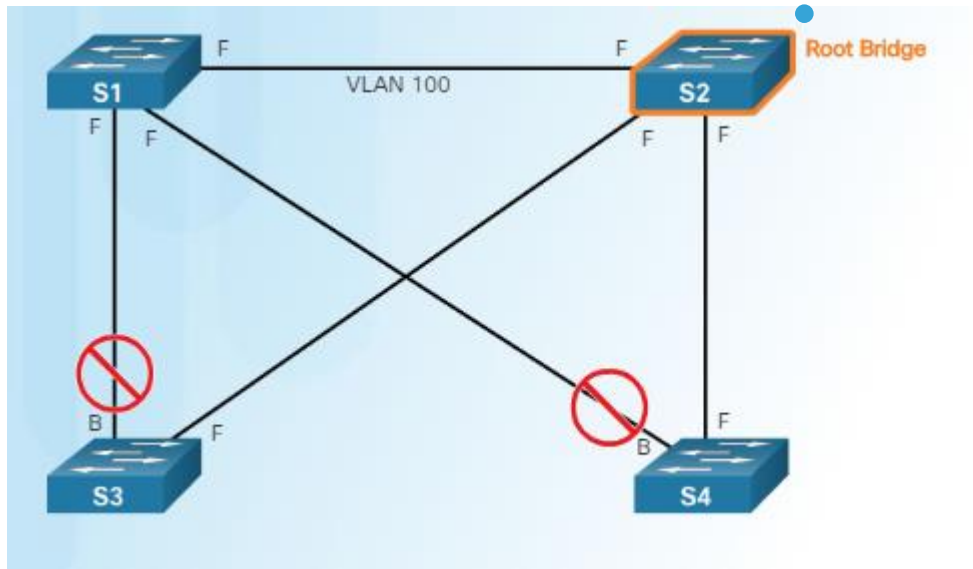
Analyzing the STP Topology



Expected Topology Versus Actual Topology

- Ensure that the spanning-tree topology matches what is expected.

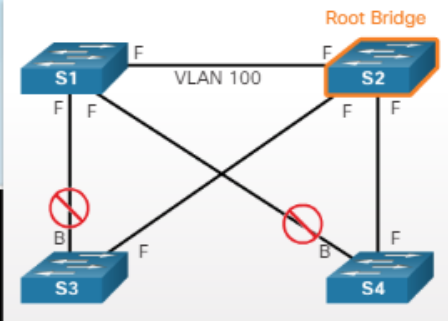
Use **show** commands to verify STP. Do not forget to verify load balancing.



STP Configuration Issues

Overview of STP Status

- Use the **show spanning-tree** and **show spanning-tree vlan x** commands to verify the STP status.



```
S1# show spanning-tree vlan 100

VLAN0100
Spanning tree enabled protocol rstp
Root ID    Priority    28772
           Address    0000.0c9f.3127
           Cost       2
           Port       88 (TenGigabit9/1)
           Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID   Priority    28772 (priority 28672 sys-id-ext 100)
           Address    0000.0cab.3724
           Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time  300

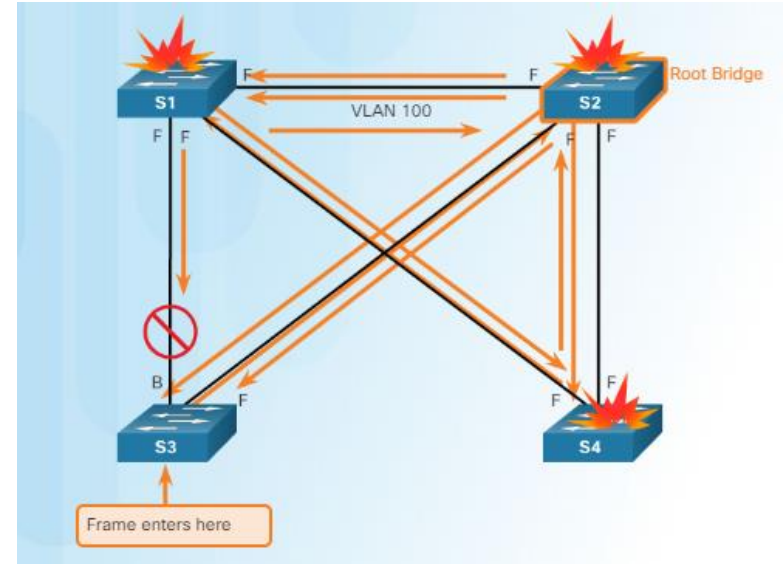
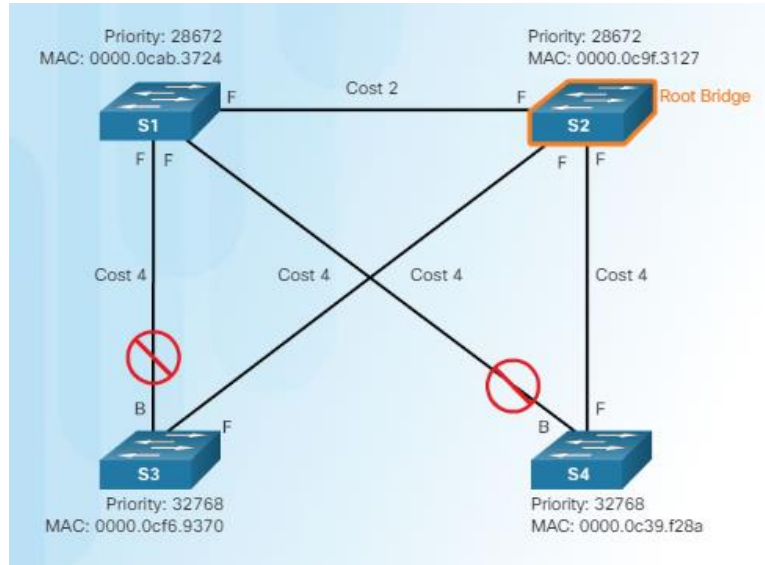
Interface   Role Sts Cost      Prio.Nbr Type
-----
Gi3/1       Desg FWD 4        128.72   P2p
Gi3/2       Desg FWD 4        128.80   P2p
Te9/1       Root FWD 2        128.88   P2p
```

Ten gigabit
Ethernet interface

STP Configuration Issues

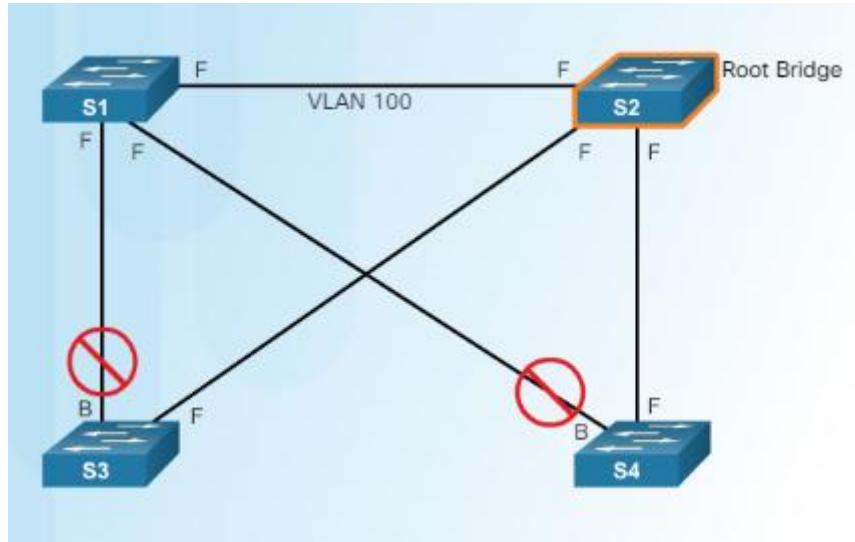
Spanning Tree Failure Consequences

- NEVER turn STP off; this can cause a switched network to be unusable – Remember that there is **not a TTL mechanism at Layer 2**.



Repairing a Spanning Tree Problem

- Manually remove redundant links (physically remove the cable OR through configuration, if possible).
 - Determine and repair the cause of the spanning tree failure.
 - If unable to determine the problem, reinstall cables one at a time (or re-enable the ports) to locate the issue.



Switch Stacking and Chassis Aggregation

Switch Stacking Concepts

- Can connect up to **nine** 3750 switches
- One switch (the **stack master**) controls the operation of the stack
 - If this switch goes down, a new stack master is elected
- Appears as one entity to the network
 - Stack is assigned **one IP** address
- Each switch has a **unique stack member number**
 - Can configure a priority value to determine which switch is stack master
 - **Highest stack member** priority value is stack master
- The stack master has the saved and running configuration files for the entire stack.
 - Only **one configuration** file to manage and maintain

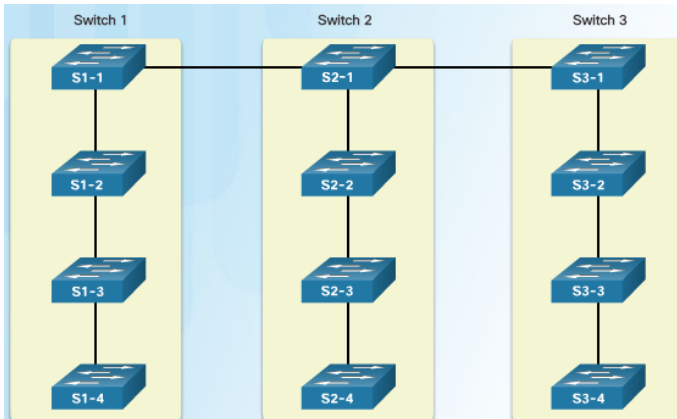
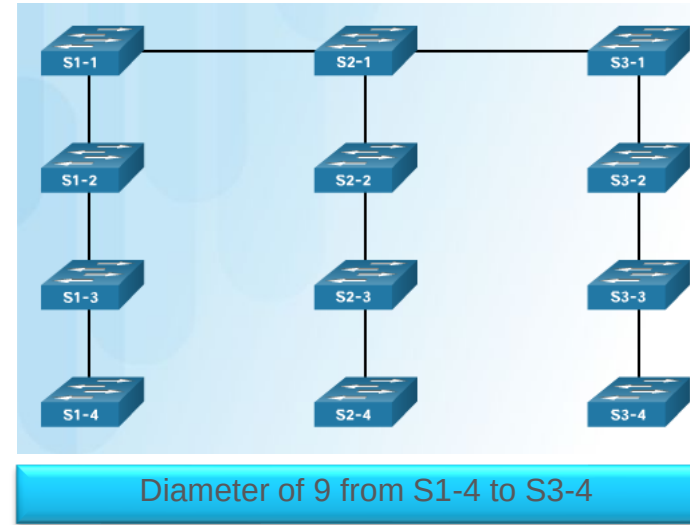


```
Switch# show running-config | begin interface
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
<output omitted>
!
interface GigabitEthernet1/0/52
!
interface GigabitEthernet2/0/1
!
interface GigabitEthernet2/0/2
!
<output omitted>
!
interface GigabitEthernet2/0/52
!
interface GigabitEthernet3/0/1
!
interface GigabitEthernet3/0/2
!
<output omitted>
!
interface GigabitEthernet3/0/52
```

Switch Stacking and Chassis Aggregation

Spanning Tree and Switch Stacks

- Each stack appears as **one spanning tree instance**
- Can add switches without affecting the STP diameter (the maximum number of switches data must cross to connect between any two switches)
 - IEEE recommends a maximum diameter of **7** switches for default STP timers
 - **Default STP timers are hello – 2 seconds, max age – 20 seconds, forward delay timer – 15 seconds**



3.4 Spanning Tree Key Points

Time To Live - TTL

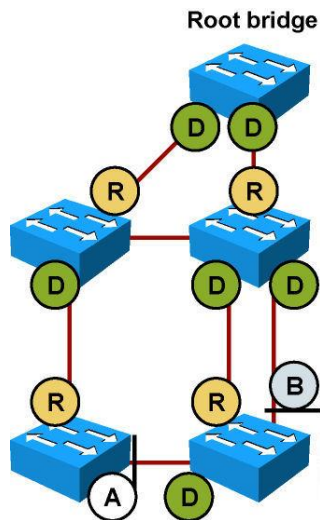
- Draw several routers connected by WAN links on the board. Draw a rectangle somewhere and tell them it is an IP header. Remind them about source and destination IP address fields. Add a field in the rectangle and label it **TTL**. Explain how **routers decrement this value each time a packet goes through the router**. When the value gets to **0**, **the packet cannot be sent any further**. This prevents packets from circling the Internet forever.
- In a separate drawing, draw several switches and draw one or more cables between them all. Do not just connect them in a straight line.
- Talk about the fact that, when data travels between **switches**, **there is no TTL-equivalent in the Ethernet Layer 2 header**. Frames that are sent that never reach their destination could theoretically **travel** between the switches **forever**.
- Write the following statement on the top of the board: **Spanning Tree Protocol creates ONE path** through the switch network. Emphasize the words **ONE** and **switch network**. Tell everyone to keep that in mind throughout the session as you explore STP.

STP Differences

- Before the lecture you may want to assign some group discussions. Have one person from each group describe the main points they discovered the day the topic is covered. Note that if you have a limited number of students, the first two bullets can be combined and the third and fourth bullets can be combined.
 - Differences between STP and PVST+
 - Differences between RSTP, Rapid PVST+, MSTP
 - Edge ports versus non-edge ports and how they affect spanning tree
 - Point-to-point links versus shared ports and how they affect spanning tree
 - Switch stacking and how it affects spanning tree

STP – Speeding up

- Talk about how **Cisco speeded up spanning tree** by providing the **PortFast** option and where to use it (ports that always have end devices attached). Industry followed by introducing **RSTP**.
- With RSTP, there are two new port states. Remind the Root Port and Designated Port states.
 - One new port state is the **Alternate Port**. An alternate port is the backup for the designated port when the other side is not a root port.
 - Another new port state is the **Backup Port**. The backup port is the backup port for the root port.



STP - Port States

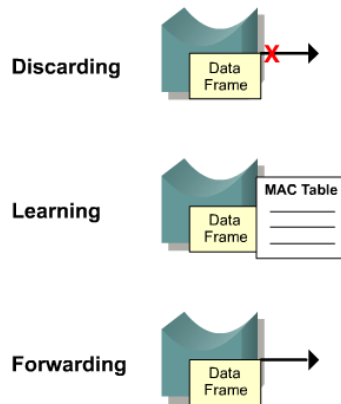
- When discussing BPDUs and the **process STP** goes through when switches **boot or a new device is connected**, remind students that the output contains information about the bridge ID (BID) or local switch information and root ID – information regarding the root bridge.
- Review the spanning tree **port states**
 - **Disabled** – spanning tree is not enabled
 - **Listening** – The port is listening to the spanning tree messages that come across as bridge protocol data units (BPDUs). The port is NOT forwarding frames.
 - **Learning** – The port is adding MAC addresses to the MAC address table (learning MAC addresses), but still not forwarding frames yet.
 - **Forwarding** – The spanning tree calculation has run and if the port is a root port or designated port, the port turns green and can send and receive data.
 - **Blocking** – The spanning tree calculation has run and because the port is not a root or designated port, the port turns amber, does not send or receive data, but it still listens for BPDU messages. Note that when a switch first turns on, all ports are in the blocking state by default.

RSTP – Port States

- Review **RSTP port states**

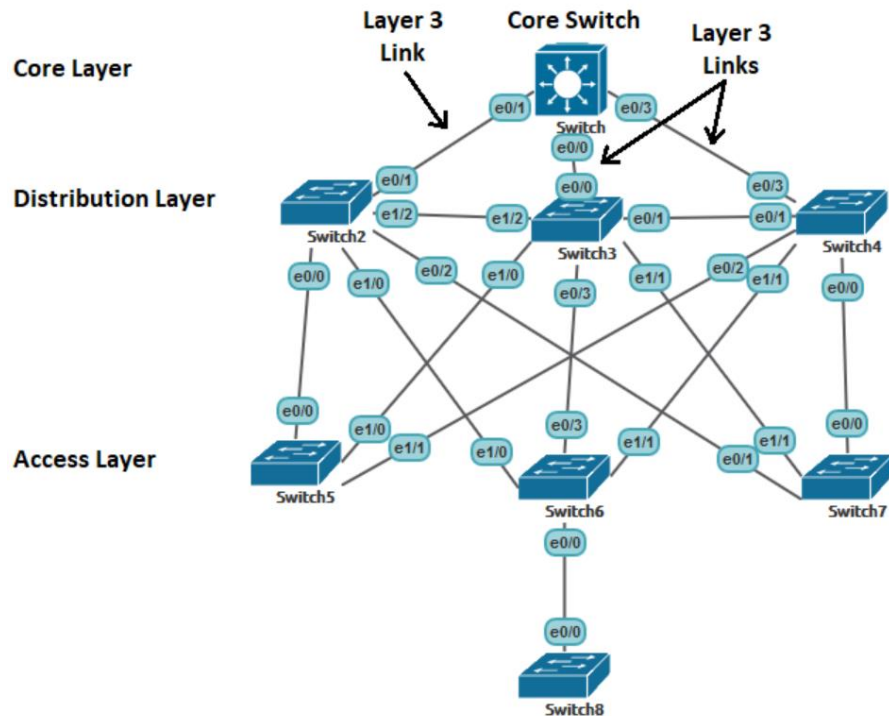
- Remember that with RSTP, there is no more disabled, blocking, or listening states. There are only the following states:
- Discarding** – no data frames are being sent out the port
- Learning** – The port is adding MAC addresses to the MAC address table (learning MAC addresses), but still not forwarding frames yet.
- Forwarding** – The spanning tree calculation has run and if the port is a root port or designated port, the port turns green and can send and receive data.

STP (802.1D) Port State	RSTP (802.1w) Port State
Disabled	Discarding
Blocking	Discarding
Listening	Discarding
Learning	Learning
Forwarding	Forwarding



STP - Issues do Occur

- Emphasize to the students that **spanning tree issues do occur**, STP is not understood well by many, and STP is on by default and should **NEVER be turned off**. If error messages occur, **troubleshoot** the cause, do NOT turn spanning tree off.



STP - Issues do Occur (Cont.)

- In order to handle STP problems, the network engineer must understand how STP operates.
- Spanning tree issues include slow traffic flow, slow remote connectivity to a switch in the affected area, and increase of dropped frames.
- Show the flowchart on 3.3.3.1. Key **troubleshooting** commands:
 - **show cdp neighbors** - see if adjacent devices can see each other. CDP is a Layer 2 technology that verifies whether one Cisco device can detect another directly connected Cisco device.
 - **show spanning-tree vlan *vlan_number*** – used to verify root bridge and port states. A technician should know what the STP states are normally, but this is not always realistic in the field.
- Techniques to **fix STP issues**:
 - **Identify** the VLAN or set of switches affected by the problem.
 - Ensure the affected **topology** is known (what specific ports are used between two switches).
 - **Disconnect redundant** links or shut the redundant ports down, if possible.
 - **Restore** disconnected redundant links **one by one** to monitor affects on spanning tree.

